

# HYUNJOON PARK

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## ABOUT ME

M.S. student at Hanyang University [HYU CVLab](#) under the supervision of [Prof. Donghyeon Cho](#). My research interests include 3D Scene Understanding by 3D Gaussian Splatting, and Conditional Generative Models.

## PATENTS AND PUBLICATIONS

C=CONFERENCE, W=WORKSHOP, U=UNDER REVIEW

- [C.2] Junha Kim, **Hyunjoon Park**, and \*Donghyeon Cho. (2026). "**High-Resolution Artwork Outpainting with Global Blueprint Guidance and Layout Control**". In *19th European Conference on Computer Vision (ECCV)*, Malmö, Sweden. [\[Project Page\]](#) [\[Paper\]](#)
- [P.1] **Hyunjoon Park**, Donghyeon Cho (2026). 다중 단서 기반 3차원 장면 이해 장치 및 방법. Korean Patent Application No. 10-2026-0105246 (Patent Pending).
- [C.1] **Hyunjoon Park**, \*Donghyeon Cho. (2026). "**Consistent Scene Understanding in 3D Gaussian Splatting via Multi-Cue Mask Refinement**". In *38th International Conference on Pattern Recognition (ICPR)*, Lyon, France. [\[Project Page\]](#) [\[Paper\]](#)
- [W.2] **Hyunjoon Park**, \*Donghyeon Cho. (2026). "**3DGS에서 2D SAM을 활용할 때 발생하는 한계점 극복 및 효과적인 3차원 공간 이해를 위한 방법에 관한 연구**". In *38th workshop on Image Processing and Image Understanding (IPIU)*, Jeju, South Korea.
- [W.1] **Hyunjoon Park**, \*Donghyeon Cho. (2025). "**3DGS를 활용한 Point Cloud 합성 및 3차원 합성 공간 복원에 대한 연구**". In *37th workshop on Image Processing and Image Understanding (IPIU)*, Jeju, South Korea.

## PROJECTS

- **Scene GraphRAG-based Video Understanding** May 2026 - Current  
Consortium: PIASPACE, Hanyang Univ., Yonsei Univ., CAU, Sejong Univ.
  - Developing a Scene Graph-based Video Understanding framework under the national IITP-GenAI research initiative.
  - Implementing GraphRAG (Retrieval-Augmented Generation) pipelines to extract multi-modal relational structures and temporal semantics from complex video sequences.
- **Conditional Video Generation: Music-to-Dance Sequence Synthesis** Feb. 2025 - Current  
Cowork: SKTelecom, Hanyang Univ. CVLab, Sungkyunkwan Univ. AIMLab
  - Developing a conditional human dance video generation pipeline built upon the U-Net framework, integrating 3D human parametric models (SMPL) and latent diffusion architectures.
  - Enhancing identity-preserving mechanism to maintain high-fidelity human characteristics, textures, and structural consistency across highly dynamic dance sequences.
  - The ultimate goal of this project is to propose a model that receives music as a condition and generates dance videos that match the music.
- **Advanced Computer Vision Final Project** Sep. 2025 - Dec. 2025  
2025 Fall Semester Advanced Computer Vision (AUE 8090, Prof. Soonmin Hwang) [\[G\]](#)
  - This project was carried out to solve the problems I felt working on the graduation project and to improve the overall performance of segmentation in the domain of 3D.
  - Addressed the critical over-segmentation and multi-view inconsistency bottlenecks of 2D foundation models (SAM) when lifting 2D masks to a 3D Gaussian Splatting scene.
  - Designed a two-stage mask refinement pipeline incorporating a depth-stratified hierarchical merging algorithm that utilizes monocular depth as both a hard geometric constraint and a soft co-planarity reward.
  - Established globally consistent 3D object identities by implementing a cross-view mask matching framework driven by agglomerative clustering of dense DINOv2 semantic features.
- **HYU Graduation Capstone Project: 3D Synthesize Space Reconstruction using 3DGS** Feb. 2025 - Nov. 2025  
Cowork: Dohyun Lim, Prof. Sungyoon Lee [\[G\]](#)
  - Developed an end-to-end 3D scene reconstruction pipeline leveraging 3D Gaussian Splatting (3DGS) to reconstruct highly realistic synthetic and real-world spaces.
  - Refined a 3D editable Gaussian Splatting framework based on Gaussian Grouping to composite distinct background and object point clouds from different source scenes into a single, unified 3D space.

## EXPERIENCE

- **CVLab @ Hanyang University**   
*Undergraduate Research Intern* Jul. 2024 - Aug. 2025  
Seoul, South Korea
  - Advisor: Prof. Donghyeon Cho
  - Thesis: 3D Editable Gaussian Splatting, Conditional Generative Model, 3D Human Reconstruction
  - Publication: "3DGS를 활용한 Point Cloud 합성 및 3차원 합성 공간 복원에 대한 연구" (IPIU, 2025)
- **LG Aimers 7th** Jul. 2025 - Aug. 2025  
*AI Hackerthon*
  - 제품 이상 여부 판별 프로젝트
- **LG Aimers 5th** Jul. 2024 - Aug. 2024  
*AI Hackerthon*
  - 식음업장 메뉴 수요 예측
- **NAVER CONNECT AI Boostcourse** Jan. 2024 - Feb. 2024  
*Online Bootcamp*
  - AI 엔지니어 기초 다지기 - 인공지능모델링

## EDUCATION

- **Hanyang University**  
*M.S. Computer Software Engineering* Seoul, South Korea
- **Hanyang University**  
*B.S. Computer Software Engineering* Seoul, South Korea
- **Hwanil High School** Seoul, South Korea

## TA EXPERIENCE

- **Creative Software Programming** Sep. 2025 - Dec. 2025  
*Dept. Computer Software Engineering, Hanyang University*
  - Assisting with lectures for weeks 13-15 and conducting practical sessions on STL (Standard Template Library)
  - Creating and grading assignments and exam questions
- **Creative Computing for Engineers** Mar. 2026 - Jun. 2026  
*Dept. Computer Software Engineering, Hanyang University*
  - Assisting with practical sessions on C and Python programming

## SKILLS

- **Programming Languages:** Python, C, C++, Java, Bash
- **Machine Learning & Vision:** PyTorch, NumPy, OpenCV, Matplotlib
- **Tools:** Git, Docker, Conda, LaTeX, Markdown

## ADDITIONAL INFORMATION

**Languages:** Korean (Native), English (Working Proficiency, TOEIC: 910), Japanese (Amateur level)

**Military Service:** Completed mandatory military service as a Sergeant and Signal Corpsman in the ROK Army (Jun. 2020 – Dec. 2021)

## REFERENCES

1. **Donghyeon Cho**  
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Department of Computer Science  
Email: doncho@hanyang.ac.kr  
*Relationship: Thesis Advisor*